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Quantum-inspired bi-level neuro-swarm optimization for UAV-based disaster recognition and response

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ABSTRACT

Reliable and timely situational awareness is essential for UAV-based disaster response yet learning-driven systems often face challenges arising from non-stationary environments, multimodal uncertainty, and limited onboard resources. The paper introduces the Quantum-Inspired Bi-Level Neuro-Swarm Optimization (QI-ENSO) method for unmanned aerial vehicle disaster relief in non-stationary and uncertain environments. The response to the calamity is treated as a bi-level problem that connects the uncertainties in perception with the decisions regarding the mission. QI-ENSO applies the learnable entanglement-based coordination and superposition-inspired exploration to reach stable convergence in the optimization landscapes characterized by non-convexity. The only algorithmic contribution made in this study is the QI-ENSO proposed method. All other aspects, like HDCA++ (the hierarchical dual-stage attention perception backbone), Digital Twin, reinforcement learning, and explainability tools, are just the non-modified and non-optimized modules that are used for evaluation and validation purposes only. Experiments conducted on multiple public UAV disaster datasets show that QI-ENSO achieves 3–7 % improvements in F1-score, faster convergence, and low edge-level inference latency (≈3.2 ms) compared to recent adaptive swarm and hybrid learning baselines. The results show that quantum-inspired coordination mechanisms provide an effective and practical optimization strategy for UAV-based disaster response under uncertainty.In this research, the latency values were measured exclusively using desktop-class GPU hardware in a controlled laboratory setting. GPU-based execution on desktop-class hardware produced all the reported inference latency results; the direct deployment on embedded or onboard UAV platforms is not included in the present study and is mentioned as future work.

1. Introduction

The world still experiences enormous human fatalities and economic losses because of natural and human-made disasters (Niquito et al., 2021). These events reveal the present technology's shortcomings, namely the lack of provision for quick and accurate awareness of the situation during complex disasters, since such events affect both the wealthiest and the poorest countries (Krichen et al., 2024). In 2024, the UAE set a record for heavy rainfall leading to severe flooding, resulting in five fatalities, major disruptions in travel, and $544 million losses (O'Connell, 2025).Experts have pointed out that the recent increase in forest fires in the Balkans has signaled the European countries' forest management practices needing very urgent strengthening(Posavec et al., 2023).The primary purpose of the forest and disaster management is to minimize the damage caused by nature's or man's intervention (Tedim et al., 2022). The pinpointing of the risk zones can, in turn, facilitate the setting up of either resistance or very lessening of the impact. Yet, in the case of the latter option, daily life could go on almost as usual with right continuous response and effective dealing with the consequences (Rezvani et al., 2023).

UAVs can greatly reduce the need for human presence at dangerous sites and help minimize errors, especially in unpredictable or hard-to-reach areas (Qin et al., 2025). UAVs can quickly spot and assess threats by reaching dangerous or hard-to-access areas and capturing

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clear, real-time images, making them highly useful for rapid situational awareness (Hussain et al., 2024) while cloud processing is powerful but unreliable during disasters due to connectivity delays and high energy demands. To overcome this, the study proposes a UAV edge framework using a transformer model for fast, on-board image processing directly at the disaster site. Moreover, trustworthy cloud services are sometimes completely off the grid (Chandran & Vipin, 2024) primarily in isolated or vulnerable areas. This situation makes it imperative to have on-board processing. However, the deployment of large vision models in drones is problematic because of their insufficient computing power and strict energy limitations(Heidari et al., 2023).Most UAV disaster management systems hardly ever apply shallow or compressed CNN models that could provide the advantages of decreased computation and energy use besides, the shallow networks frequently overlook the features essential for right classification, while the regular CNNs are perplexed by the limitations of the receptive field and thus, unable to grasp the larger, wider patterns (Mohsan et al., 2023; McEnroe et al., 2022). This challenge calls for more advanced models like transformers (Xiao et al., 2025). The study uses such complex networks to improve on-board disaster classification for UAVs. We develop a UAV-assisted edge framework that trains a transformer-based model on multiple aerial disaster datasets to achieve robust onboard classification (Jankovic et al., 2025). This network is then fine-tuned to maximize accuracy and performance while operating on edge devices (Javaid et al., 2024).The research takes on the challenge of inadequate datasets that mainly contain insufficient occurrence or concentrate on one type of disaster, hence, their usability in real-world applications is diminished. Using both aerial footages shot by UAVs and ground-level photos shot by people present, we compile seven unique catastrophe datasets (Rahnemoonfar et al., 2023).

Fig. 1. shows the UAV Model for Disaster Response.

Adequate catastrophe management and recovery become much more complicated in hazardous environments and hard-to-reach terrain Rahnemoonfar et al. (2023). UAVs with built-in platforms and optical sensors can help address these challenges (Pi et al., 2021). The work presented in this paper is aimed at making it possible to interpret aerial images on the spot and thus ensure rapid and precise hazard detection in real time(Sirma et al., 2025), (Rahnemoonfar et al., 2023).UAVs struggle in these settings because of their limited hardware, while issues like privacy, connectivity, and latency add further challenges—but many of these problems can be reduced with better system design.Using our suggested model that optimizes onboard real-time aerial picture identification, we present a UAV-assisted edge system for catastrophe detection (Rahnemoonfar et al., 2021). Considering their constrained hardware and energy provisioning, large neural networks do not lend themselves well to being run aboard UAVs. High memory use and added latency make real-time vision tasks even more difficult (Weber et al., 2022; Rahnemoonfar et al., 2023). Compressed and shallow CNN models are rarely used in UAV disaster systems because they lack the complexity needed to learn strong, useful features for correct classification (Munawar et al., 2021). Conversely, bigger receptive fields are a challenge for CNNs. As a result, we must think of more intricate designs like transformers (Bashir et al., 2024; Munawar et al., 2021). Convolutional neural networks (CNNs) address the very pressing issue of the on-board disaster category of UAVs (Khan et al., 2022). This study proposes a UAV edge framework that trains a transformer model on multiple aerial datasets and fine-tunes it to balance accuracy and speed, making it suitable for real-time use on resource-limited UAVs (Khan et al., 2022; Bashir et al., 2024).We also tackle the problem that existing datasets are too limited, often covering only a few disaster types—making them less suitable for real-world deployment. Merging information from seven different disaster scenarios, which were documented by both on-ground staff and UAVs, enables us to create an exceptional database (Hussain et al., 2024; Khan et al., 2022).

The novelty of this work lies in a quantum-inspired bi-level neuro-swarm optimization formulation in which a learnable entanglement matrix explicitly couples perception uncertainty with UAV decision variables. Unlike classical swarm or evolutionary methods, this formulation enables coordinated exploration and stable convergence under non-stationary disaster environments. Supporting contributions include: (i) A hierarchical dual-stage attention network (HDCA++) used solely to generate stable multimodal features for optimizer evaluation, (ii) A digital twin environment employed for safe policy rehearsal and stress testing and(ii) Explainability tools used for post-hoc validation, not algorithmic optimization.

Even though the experimental pipeline combines perception, simulation, reinforcement learning, and explainability, this paper delivers only one algorithmic contribution: the Quantum-Inspired Bi-Level Neuro-Swarm Optimization (QI-ENSO) paradigm. The remaining parts like HDCA++, Digital Twin–based policy rehearsal, reinforcement learning agents, and post-hoc explainability tools are all considered as fixed evaluation modules. These components are only used to evaluate the robustness, stability, and practical effectiveness of QI-ENSO in the context of realistic UAV disaster-response scenarios.

Apart from the methodological innovation, particular stress is laid on reproductive capabilities. An exhaustive algorithmic specification is given, which includes direct update equations for all the learnable parts and a thorough computational complexity analysis. These factors combine to guarantee that the suggested optimization process can be executed and verified by others without the support of hidden assumptions or secret procedures thus making it all truly clear.

This work makes the following contributions:

- **Quantum-Inspired Bi-Level Optimization Formulation**

The development of a quantum-inspired bi-level neuro-swarm decision optimization framework is proposed, wherein perception uncertainty is skillfully interwoven with UAV decision variables in a common learning-optimization loop. Disaster management is structured as a layered optimization problem, where strategic intent at the top level and control policies at the lower level have simultaneously evolved by means of intertwined neuro-swarm dynamics. The framework, using quantum-inspired exploration methods and bi-level feedback, effectively reduces the occurrences of instability, stagnation, and premature convergence, which are drawbacks that often plague traditional swarm intelligence and reinforcement learning techniques. The new system allows for the use of strong, flexible, and uncertainty-aware decision-making processes in extremely mobile and only partially visible disaster settings.

- **Entanglement-Driven Particle Coordination Mechanism**

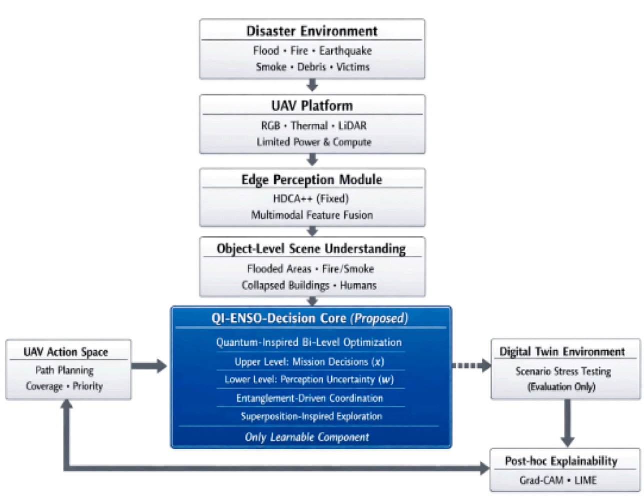

Fig. 1. Disaster Scenario driven UAV decision disaster response framework.

A learnable entanglement matrix is introduced to enable adaptive, state-dependent coupling among swarm particles. The proposed mechanism is different from the correlation-based methods and fixed-interaction models since it makes the exploration and refinement processes carried out simultaneously and adapt to the changes in the optimization landscape.

- **Superposition-Inspired Multi-Directional Search Strategy**

The optimizer includes a superposition-motivated exploration mechanism that checks various applicant directions at the same time, global search coverage is enhanced while at the same time stability of the non-stationary disaster environments convergence is preserved.

- **Comprehensive Evaluation under Realistic UAV Conditions**

The suggested optimizer undergoes a thorough assessment in conjunction with a multimodal perception network, Digital Twin-based policy rehearsal, and post-hoc analysis of the explanations as support. The extensive testing carried out on different drones' disaster datasets indicates that the new technique not only outperforms the old one but also is very dependable and has a perfect fit for real-time edge deployment.

The remaining parts of the document are structured in the sequence of the background and relevant literature in Section 2, the Proposed Methodology in Section 3, and Results and Discussion in Section 4. Some concluding thoughts follow in Section 5 and then the future research areas are mentioned.

2. Related work

The search capabilities based on the population together with the robustness against local optima have led to an extensive investigation of swarm intelligence algorithms for the resolution of non-convex and dynamic optimization problems. Particle Swarm Optimization (PSO) and Differential Evolution (DE) are among the traditional techniques that have predefined interaction rules and fixed communication structures among agents, which could limit their suitability for rapidly changing environments. Recent advances have proposed the use of swarm optimizers that dynamically control parameters like inertia weights, learning coefficients or mutation strategies during the search process to mitigate these drawbacks. Improvements in convergence stability accompany these methods, but the interaction topology between agents continues to be largely fixed and non-learnable thus limiting their ability to act together when the situation is not clear (Pan et al., 2021; Xie et al., 2024).

More recent work has explored cooperative and multi-swarm frameworks, where the population is partitioned into sub-swarms to enhance diversity and mitigate premature convergence (Gao et al., 2024). Although effective in improving exploration, these methods rely on heuristic information exchange mechanisms and do not explicitly learn inter-agent dependencies. As a result, coordination appears indirectly rather than being optimized as part of the search dynamics.

2.1. Hybrid swarm–reinforcement learning approaches

In order to enhance the flexibility of the systems to cope with the changes, many researchers have explored the application of the combination of swarm optimization and Reinforcement Learning(RL).RL is commonly used in these combined methods to adjust swarm parameters or to help select actions, which in turn leads to more efficient exploration-exploitation trade-offs (Yin et al., 2023; Wang et al., 2022). One drawback that most of the hybrid swarm-RL methods have been that they consider learning as a secondary process, thus, the underlying swarm interaction structure continues to be fixed. As a result of these methods, control policies are improved but the agents' influence on each other during optimization is not fundamentally changed.

2.2. Quantum-inspired optimization methods

The innovation of Quantum-inspired optimization has offered a way forward for enhancing the global search quality by applying the principles of superposition, probabilistic state representation, and population diversity, among others. Quantum-inspired evolution and swarm optimizers have shown and claimed better convergence and exploration performance than the classical versions (Wang & Wang, 2021; Pathak et al., 2022; Manfreda et al., 2019). However, existing quantum-inspired methods typically apply quantum concepts at the parameter or representation level and do not explicitly model learnable inter-agent coupling. Their adaptive learning to cope with various dynamic real-world conditions and phases of the learning landscape is disabled.

2.3. Optimization for UAV-based disaster response

In UAV-based disaster response, optimization plays a critical role in perception robustness, decision-making, and resource allocation under uncertainty. Swarm intelligence and learning-based optimization have been used in the monitoring of disasters, the planning of paths and the assessment of damage with the help of aerial images in the past years (Kerle et al., 2019; Yazdani et al., 2023). Although these methods have massive potential in the application area, they still work with traditional optimizers. Such methods never clearly deal with uncertainty-aware coordination or interaction learning. This discrepancy is most prominent in dynamic disaster in which the conditions of the sensors are rapidly changing in real-time. The challenge needs the use of more high-tech models, such as transformers (Xiao et al., 2025). Since recently, the transformer type of vision models has proved to be more competent in global context modeling than convolutional architecture, thus teaching representation of complex visual scenes better (Dosovitskiy et al., 2021).

2.4. Research gap and motivation

The above review clearly reveals that existing swarm-based, hybrid learning, and quantum-inspired optimization methods predominantly rely on parameter adaptation or population partitioning, while explicit learning of inter-agent interactions stays largely unaddressed. Moreover, hardly any research has been done that treats UAV-based disaster relief as a bi-level optimization problem in which perception uncertainty is linked to the mission-level decision-making process. Therefore, the existing methods are characterized by a limited degree of adaptability and coordination when they function in the very dynamic and uncertain environments brought about by disasters.

Recent research indicates that interaction-inspired optimization mechanisms can significantly improve UAV-based perception, especially in resource-limited situations and when dealing with small or distant targets. However, these techniques generally regard perception, optimization, and decision-making as individual processes. Such a disconnected framework very much restricts the scope for coordinated adaptation, which in turn lessens the system's resilience in real-world disaster-response scenarios.

To cope with these deficiencies, the suggested QIBNS++ framework not only takes a unified optimization paradigm but also combines quantum-inspired swarm intelligence with uncertainty-aware decision-making. The framework, by explicitly learning adaptive inter-agent coordination and dynamically balancing exploration and exploitation within a bi-level formulation, yields better robustness, interpretability, and operational responsiveness. The systematic and integrated approach does not only concentrate on the intermediate improvements but also sets up a more reliable and consistent basis for the UAV-mediated disaster management in the complex and real-world environments.

3. Materials and methods

3.1. Overall proposed framework

The total experimental setup is to ensure that the Quantum-Inspired Bi-Level Neuro-Swarm Optimization (QI-ENSO) process for disaster recognition and response by UAVs gets a fair, transparent and reproducible evaluation. The framework utilizes an extremely modular plan with a clear separation of the optimizer from the hardware, thus making it possible to assign any increase in performance solely to QI-ENSO. The experimentation procedure is divided into four stages, which follow one another: recognition of the feature, bi-level optimization, policy rehearsal, and hindsight validation. First, a multimodal UAV sensory input is fed through a fixed perception backbone (HDCA++) which stabilizes feature representation. This perception module is consistent throughout all tests and is not changed, trained, or optimized for the purpose of this study. Then, the QI-ENSO optimizer, the only adaptive and learnable component in the system, receives the extracted features. QI-ENSO conducts bi-level optimization by connecting perception-level uncertainty with mission-level decision variables via entanglement-driven coordination and superposition-inspired exploration. The decision policies that come out of this are tested in a Digital Twin setting, which mimics different disaster scenarios, sensor noise, and changes in the environment. The Digital Twin is solely responsible for policy rehearsal and robustness testing and does not add any learning or optimization.

Ultimately, explainability tools like Grad-CAM and LIME are used post-hoc to investigate decision behavior and model consistency. These tools are applied exclusively for interpretability and qualitative validation purposes and do not participate in optimization. The same datasets, preprocessing methods, data splits, and evaluation metrics are applied during all the experiments. This common framework provides methodological rigor, prevents the results by the auxiliary modules, and confirms that the reported improvements in performance, stability, and convergence come solely from the new QI-ENSO optimization strategy. Fig. 2 shows the framework architecture for evaluation of UAV disaster-response.

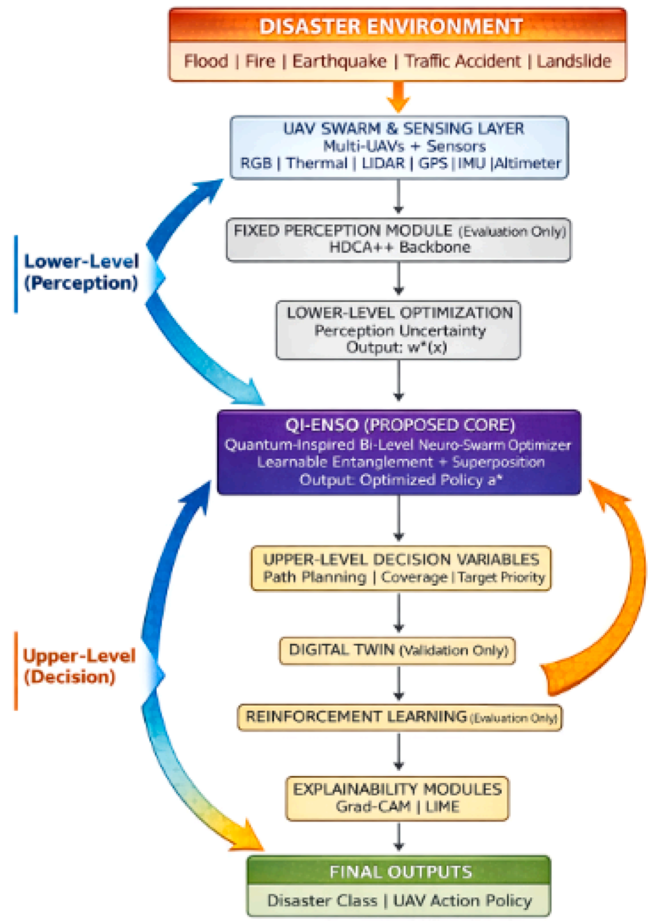

Fig. 2. Architecture for evaluation of UAV disaster-response with QI-ENSO as the only optimization module highlighted and perception, simulation, learning, and explainability as the constant evaluation elements.

3.2. Proposed methodology

In this section, the suggested Quantum-Inspired Bi-Level Neuro-Swarm Optimization (QI-ENSO) for UAV-based disaster recognition and response optimization framework is explained, which is the only contribution of this research in terms of method and algorithms. The method is deliberately designed to separate the optimizer from all the supportive components. Perception (HDCA++), Digital Twin simulation, reinforcement learning, and explainability tools are considered as fixed, standard modules and are used only for evaluation, stress testing, and validation. No learning, optimization, or architectural novelty is claimed for these supporting components.

The classical swarm optimizers like PSO, ABC, and DE usually depend on weak or fixed interaction structures, typically directed by the global or the best local solution. Such methods in very non-stationary disaster environments often lead to premature convergence, oscillation, or unstable learning. The new method offers a coupled swarm coordination mechanism to manage the swarm agents. QI-ENSO puts forward two quantum-inspired principles such as entanglement and superposition that are converted into a mathematically realizable optimization framework to tackle these limitations. QI ENSO is a classical optimizer that does not simulate quantum physics but rather utilizes the properties of the quantum system like coordinated state dependency and parallel exploration to enhance the efficiency of the search and the quality of decisions made

3.2.1. Problem formulation and bi-level optimization objective

The problem of UAV disaster response is formulated as a bi-level optimization framework where the decisions at the mission level are optimized by taking the behavior of the perception module as best completely. UAV that supports disaster response on ground through aerial view requires two levels of optimization that are tightly coupled:

- **Lower level (perception optimization):** learning stable and discriminative representations from noisy, multimodal sensory inputs under resource constraints.
- **Upper level (decision optimization):** selecting UAV actions (navigation, prioritization, coverage) that maximize mission effectiveness under uncertainty.

At the lower level, perception parameters are learned under practical constraints such as limited resources and sensor uncertainty, while the upper level focuses on selecting safe, energy-efficient, and well-covered UAV actions. The interaction between decision quality and perception uncertainty is therefore directly perceived through the proposed model which combines the two levels together. This interaction, which is usually ignored in the case of single-level optimization or traditional reinforcement learning approaches, is made possible to be evaluated based on the quality of the decision made under the given uncertainty level. To have precise and open appraisal, all parts of the system are kept the same for all the tests, and improvements in the performance seen are only linked to the suggested bi-level optimization technique.

An optimization problem with bi-level interaction is structured as subordinated decision variables as shown below:

Upper-Level Problem

$$\begin{array}{ll} \min\limits_{\mathbf{x}} & F(\mathbf{x}, \mathbf{w}^*(\mathbf{x})) \\ \text{s.t.} & \mathbf{w}^*(\mathbf{x}) = \underset{\mathbf{w}}{\operatorname{argmin}} | f(\mathbf{w}, \mathbf{x}), \\ & g_u(\mathbf{x}) \leq 0, \\ & h_u(\mathbf{x}) = 0, \end{array} \tag{1}$$

Lower-Level Problem

$$\begin{array}{ll} \min\limits_{\mathbf{w}} & f(\mathbf{w}, \mathbf{x}) \\ \text{s.t.} & g_l(\mathbf{w}, \mathbf{x}) \leq 0, \\ & h_l(\mathbf{w}, \mathbf{x}) = 0. \end{array} \tag{2}$$

This hierarchical setup without any doubt reveals the dependence of mission-level decision making on perception-level optimization, which cannot be represented using a one-level formulation.

3.2.2. Entanglement-driven particle coupling

Intuitively, the entanglement mechanism can be viewed as a soft, adaptive communication layer among swarm agents. Rather than following fixed neighbors or global best solutions, particles dynamically adjust their influence on one another based on optimization feedback, allowing stronger coordination in uncertain regions and looser coupling during broader exploration. This behavior resembles cooperative adaptation seen in human or biological teams operating under changing conditions.

Let $\mathrm{x_i^t} \in \mathrm{R^D}$ denote the state of particle i at iteration t. QI-ENSO introduces a learnable entanglement matrix $\mathrm{E_q} \in \mathrm{R^P}$.

The particle updates are governed by the rule of interaction stated in (3), and the calculation is done as follows.

$$\mathrm{X_i}^{\mathrm{t+1}} = \mathrm{x_i}^{\mathrm{t}} + \alpha \sum \mathrm{E_q(i,j)} \cdot \left(\mathrm{x_j}^{\mathrm{t}} - \mathrm{x_i}^{\mathrm{t}}\right) \tag{3}$$

where in $\mathrm{Eq(i,j)}$ indicates the adaptive influence strength regulating the connection between particles i and j, and the coupling coefficient α determines the total power of information transfer through the swarm. This updating system lets the particles modify their paths in accordance with the group movement, remaining stable and flexible at the same time during the optimization process.

The update of the entanglement matrix is entirely predictable, and it is solely the optimization feedback that regulates it. The entanglement coefficients are modified at each iteration according to the relative enhancement of the bi-level objective, which allows for the adaptive inter-particle coupling and reproducibility at the same time. The update procedure does not require any dataset-specific tuning, stochastic heuristics, or implicit rules, which leads to reliable and uniform performance among different implementations.

Classical PSO depends only on individual and global best attraction, while ABC relies on probabilistic role switching. Neither provides structured, learnable inter-agent coupling.

3.2.3. Superposition-inspired multi-directional exploration

To avoid stagnation, QI-ENSO augments entanglement with a **superposition-inspired exploration step**, allowing particles to evaluate multiple candidate directions simultaneously:

$$\mathrm{X_i}^{\mathrm{t+1}} \leftarrow \mathrm{x_i}^{\mathrm{t+1}} + \beta \cdot \Delta_{\mathrm{i}}(\mathrm{k}) \tag{4}$$

where $\Delta_{\mathrm{i}}(\mathrm{k})$ represents k randomly sampled directional perturbations and β controls exploration magnitude.

The mechanism expands the search space without random walks, and the controlled parallel exploration becomes possible. Combined with entanglement, it balances global exploration and local refinement.

The QI-ENSO framework proposed here has been described using a fully specified step-by-step algorithm to support reproducibility and independent verification. All update rules, including the adaptive entanglement-driven particle coupling mechanism, are presented mathematically in explicit terms. Heuristic steps or assumptions related to specific implementations are not kept under wraps. In this sense, Algorithm 1 gives a complete procedural description that is adequate for direct re-implementation. All initialization steps, updating rules, stopping criteria, and parameter dependencies necessary for the reproduction of the suggested QI-ENSO optimizer are defined in Algorithm 1. There is no need for any outside procedures or undisclosed implementation details.

Algorithm 1 illustrates the proposed QI-ENSO – Quantum-Inspired Bi-Level Neuro-Swarm Optimization algorithm.

Algorithm 1. Quantum-Inspired Bi-Level Neuro-Swarm Optimization (QI-ENSO)

Objective: Optimize the recognition and response decisions during disasters by UAVs through the utilization of a quantum-inspired bi-level swarm optimization framework which made explicit the coupling of perception uncertainty with mission-level decision variables.

Remark: The HDCA++ perception backbone, Digital Twin simulation environment, reinforcement learning baselines, and explainability modules are not included in the optimization process. The components are constant and are merely used for the purposes of post-optimization evaluation and performance measurement.

Input: Multimodal UAV observations $\mathscr{M}$; frozen perception backbone (HDCA++); Digital Twin environment $\mathscr{D}$; swarm size P; maximum iterations T; entanglement coefficient α; superposition coefficient β.

Output: Predicted disaster class $\widehat{Y}$; optimized UAV action policy $\mathbf{a}^*$.

1. **Perception Encoding (Fixed):**
   Extract and fuse multimodal features using the frozen HDCA++ backbone to obtain $\mathbf{F}_{\text{fused}}$.
2. **Initialization:**
   Initialize particle swarm $\{\mathbf{x}_i^0\}_{i=1}^P$ encoding perception-aware decision variables; initialize entanglement matrix $\mathbf{E}_q \in \mathbb{R}^{P\times P}$.
3. **for** $t = 0$ **to** $T-1$ **do**
   - **Bi-level Evaluation:**
     For each particle $\mathbf{x}_i^t$, solve the lower-level objective $f(\mathbf{w}, \mathbf{x}_i^t)$and evaluate the upper-level objective $F(\mathbf{x}_i^t, \mathbf{w}^*(\mathbf{x}_i^t))$.
   - **Entanglement Update:**
     Update particle positions via adaptive coupling:
     $$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \alpha \sum_{j=1}^{P} E_q(i,j) \left(\mathbf{x}_j^t \mathbf{x}_i^t\right)$$
   - **Superposition Exploration:**
     Sample candidate perturbations $\{\Delta_i^{(k)}\}$, select the most promising direction, and update:
     $$\mathbf{x}_i^{t+1} \leftarrow \mathbf{x}_i^{t+1} + \beta \Delta_i^{(k)}$$
   - **Entanglement Adaptation:**
     Update $\mathbf{E}_q$using optimization feedback and uncertainty sensitivity.
4. **end for**
5. **Validation (Evaluation Only):**
   Validate the optimized solution in the Digital Twin environment $\mathscr{D}$ to obtain $\mathbf{a}^*$.
6. **Interpretability (post-hoc):**
   Apply Grad-CAM and LIME for explanation without influencing optimization.

Return: $\widehat{Y}, \mathbf{a}^*$.

3.2.4. Notations used in the proposed QI-ENSO framework

To enhance the understanding and uniformity of mathematical representation and algorithmic explanation, the major symbols and notations utilized all along the suggested QI-ENSO framework are listed in Table 1. The same notations are used in a consistent manner throughout the bi-level optimization model, swarm interaction mechanisms, and experimental evaluation parts.

3.2.5. Computational complexity

The computational cost of the proposed QI-ENSO framework is characterized by the operations associated with its main algorithmic components. Let P indicate the number of particles in the swarm, $d$the dimensionality of the decision space, K the number of candidate perturbation directions used during superposition-based exploration, and $T$the total number of optimization iterations.

During each iteration, every particle undergoes a bi-level objective evaluation. This involves solving a lower-level optimization problem with computational cost $\mathscr{O}(C_f)$, followed by an upper-level objective evaluation with cost $\mathscr{O}(C_F)$. As these evaluations are performed independently across all particles, the resulting per-iteration cost is

$$\mathscr{O}\left(P\left(C_f + C_F\right)\right) \tag{5}$$

The entanglement-driven coordination stage requires fully connected interactions among particles, which leads to a computational cost of $\mathscr{O}(P^2 d)$per iteration. Additionally, the superposition-inspired exploration mechanism evaluates Kperturbation candidates for each particle, contributing an extra $\mathscr{O}(PKd)$cost per iteration. Updating the entanglement matrix $\mathrm{E}_q \in \mathbb{R}^{P \times P}$ further introduces a computational overhead of $\mathscr{O}(P^2)$at each iteration.

Combining these terms, the overall time complexity of QI-ENSO can be expressed as

$$\mathscr{O}\left(T\left[P\left(C_f + C_F\right) + P^2 d + PKd\right]\right), \tag{6}$$

with the entanglement-driven coordination term $\mathscr{O}(TP^2 d)$ dominating for large swarm sizes.

In terms of memory requirements, the algorithm has a space complexity of $\mathscr{O}(Pd + P^2)$, which accounts for storing the particle states and the entanglement matrix. This complexity analysis underscores the inherent trade-off between rich inter-particle coordination and scalability in population-based optimization methods, particularly in uncertain and high-dimensional environments.

The inter-agent interactions, however, are the computational overhead that the proposed design mainly aimed at small-to-medium-sized UAV swarms, which, due to practical constraints such as airspace safety, communication bandwidth, and onboard energy limits, are most deployed in real-world disaster-response missions. To further improve scalability, the entanglement matrix can be made sparse or localized, thus effectively reducing the dense global interactions to neighborhood-level coordination. Such extensions enable decentralization and scalable implementations and are therefore identified as natural directions for future work rather than limitations of the current framework. Importantly, within the swarm sizes evaluated in this study, QI-ENSO consistently shows stable convergence and low inference latency, confirming its suitability for practical and operational deployment.

The comprehensive analysis of computational complexity done earlier made it possible to set specific limits for every phase of the QI-ENSO optimization process. It is the dependence on the size of the swarm, the dimensionality of the solution, and the number of iterations that have been indicated that allows practitioners to not only replicate the performance behavior but also the runtime characteristics under the same experimental conditions.

Table 1
Notations used in the proposed QI-ENSO framework.

Notation	Description
x	Upper-level (mission-level) decision variables
w	Lower-level (perception-level) variables
F(x,w*(x))	Upper-level bi-level objective (mission effectiveness)
f(w,x)	Lower-level perception optimization objective
$g_u(\cdot), h_u(\cdot)$	Upper-level inequality and equality constraints
$g_l(\cdot), h_l(\cdot)$	Lower-level inequality and equality constraints
P	Number of swarm particles
D	Dimensionality of solution space
T	Maximum number of iterations
x_i^t	Position of particle $i$at iteration t
A	Entanglement-driven coupling coefficient
B	Superposition-based exploration coefficient
$Eq \in R^{P \times P}$	Learnable entanglement matrix
$\Delta_i^{(k)}$	k-th candidate perturbation direction
$\hat{Y}$	Predicted disaster class
a*	Optimized UAV action policy
Tp, Tn, Fp, Fn	True positives, True negatives, False Positives, False Negatives

3.3. Datasets description

For the sake of experimental transparency and reproducibility, comprehensive information about the sizes of the datasets, the distributions of the classes, and the data splits is given. Every dataset is accessible to the public and has undergone the same preprocessing and partitioning protocol in all the experiments and were partitioned using identical train/validation/test splits. The application of this unified protocol guarantees an equal evaluation and reproduction of results throughout the entire series of experiments.

This study employs four publicly available datasets commonly used for UAV-based disaster analysis-AIDER, C2A, DisasterEye, and RescueNet datasets with all detailed statistics as a basis for reproducibility and transparent evaluation. The AIDER dataset consists of >6000 aerial images, which are categorized into five classes: Collapsed Building, Fire/Smoke, Flood, Traffic Accidents, and Normal, and this leads to an imbalanced classification setting. The C2A dataset has 10,215 high-resolution images depicting four disaster scene types (fire/smoke, flood, collapsed building/rubble, and traffic accident) plus human pose annotations in five pose categories. The RescueNet dataset has 4494 high-resolution images taken after the disaster and divided into 3595 for training, 449 for validation, and 450 for testing samples, each annotated with one of the 11 classes: background, water, vehicles, and building damage levels. For the DisasterEye dataset, we utilize the publicly available version presented in the original paper since there are some differences between releases; therefore, we keep the original class definitions and splits provided by the dataset authors. The statistics and compositions of these datasets guarantee the transparency of experiments and the fairness of comparisons between different models.

The chosen data sets are a combination of different disaster situations, signal quality and class frequencies, the latter being the one widely present in UAV-based disaster response. The use of several public datasets with different degrees of difficulty not only presents a chance to assess the system's generalization and robustness but also to reproduce the results and make direct comparisons with recent research done in the same area. Together, they provide a balanced testbed for assessing optimization behavior under realistic operational variability.

3.4. Data preprocessing

All the images were subjected to a universal preprocessing pipeline which made the datasets consistent and prevented any introduction of bias that was specific to a dataset. The images were downscaled to a uniform resolution and then normalized according to the mean and standard deviation values of the ImageNet protocol. Normalization leads to steady optimization and uniform training behavior across models.

To strengthen the generalizing ability of the model, data augmentation was restricted to the training set only. The augmentation methods included random flipping both horizontally and vertically, rotating, changing the size, and changing the colors. Same augmentation techniques and parameter settings were applied for all datasets and models to ensure proper comparison.

The same preprocessing steps were taken for both conventional methods and the proposed system which made it clear that the performance differences observed in the experiment were due to the model's architecture and optimization strategies, not the input preprocessing variations.

3.5. Theoretical comparison with recent swarm-based optimizers

Recent swarm-based optimization methods aim to improve convergence and balance exploration and exploitation in complex, non-convex problems. Adaptive PSO variants achieve this mainly through dynamic adjustment of inertia and learning parameters, while cooperative and multi-swarm DE frameworks enhance diversity by dividing the population into subgroups. More recent hybrid swarm–learning and opposition-based approaches further expand global exploration; however, these methods typically rely on fixed or heuristically defined interaction structures.

Although effective in moderately dynamic settings, such approaches keep a predefined communication topology. The coordination between agents is not direct but rather through their attraction to the global or best neighborhood solution which is the primary way of learning inter-agent relationships. Consequently, interaction strength stays static and does not adapt to changing optimization conditions.

On the other hand, QI-ENSO takes a completely different approach by making interaction learning a part of the optimization process. A learnable entanglement matrix that handles explicitly modeling state-dependent coupling among the swarm particles is not only created but also updated while optimization. It is almost like this that the interaction topology is allowed to change along with the search landscape, and as a result a swarm of dynamically self-organizing agents is created.

The entanglement in QI-ENSO mechanism allows particles to aid each other more when the uncertainty is at its peak and to do it less when the exploration is going on. Such behavior of optimization is not dictated by fixed interaction rules but is the result of the process. Moreover, the superposition-inspired exploration strategy allows particles to assess several update directions simultaneously, thus, facilitating structured exploration that eludes the instability often associated with random or opposition-based perturbations.

Another point of differentiation is the bi-level optimization formulation of QI-ENSO, which visibly merges the perception uncertainty with the mission-level decision variables. This method allows the uncertainty-aware feedback to get involved in the decision-making process, unlike single-level swarm optimizers that integrate all objectives into one cost function, straight away. Typically, QI-ENSO is a technique that not just modifies parameters, but also understands communication, cooperation, and different areas of optimization that agents are using. This adaptive interaction learning is especially compatible with UAV-based disaster response, where changing environments and uncertain sensing require real-time reorganization rather than fixed coordination rules.

Lately, hybrid swarm – reinforcement learning techniques have relied mostly on RL not just for adapting the swarm parameters but also for selecting the actions to be taken, while the basic interaction of the agents has been either fixed or governed by some heuristics. QI-ENSO, on the other hand, regards the coordination of agents as a parameter that can be learned and controlled through an entanglement-driven coupling mechanism that adapts to the changes of the optimization landscape. The system allows the coordination to be adaptive under uncertainty without the need for integrating a complete reinforcement learning loop into the optimizer, so achieving better stability and lower training complexity for the real-time UAV applications. Key structural differences in interaction mechanisms, adaptability, objective formulation, and convergence behavior are summarized in Table 2.

Although formal convergence proofs are intractable for non-convex swarm optimizers, QI-ENSO empirically exhibits monotonic convergence trends with reduced oscillation.

3.6. Perception module (HDCA++)

To evaluate QI-ENSO under realistic perception uncertainty, we employ HDCA++, a hierarchical dual-stage cross-attention network, strictly as a fixed perception backbone for generating stable multimodal features. HDCA++ is not proposed as a novel method in this work; it is used solely to support the evaluation of the proposed QI-ENSO optimizer.

HDCA++ decouples:

- **Intra-modal refinement** via self-attention, and
- **Inter-modal alignment** via bidirectional cross-attention.

This design ensures stable multimodal representations but does not constitute the primary contribution. All perception models compared in experiments use identical feature extractors to ensure that observed performance gains originate from the optimizer.

3.7. Digital Twin–driven policy rehearsal

The Digital Twin is employed exclusively for policy rehearsal and robustness testing and does not constitute an algorithmic contribution.

The QI-ENSO model, using a digital twin (DT), is subjected to controlled environmental variability exclusively for the purpose of experimenting with decision policies in a safe manner and testing the extent of their strength under the circumstances of sensor failures and changing hazard dynamics.

The DT does not introduce new learning algorithms; it serves as a stress-testing environment to validate the optimizer's adaptability under non-stationary conditions.

3.8. Explainability as post-hoc validation

Explainable AI tools, specifically Grad-CAM and LIME, are applied

Table 2
Structural comparison with recent swarm optimizers.

Optimizer	Interaction Structure	Adaptivity	Objective Level	Convergence Stability
Adaptive PSO (Yazdani et al., 2023)	Global / local best attraction	Parameter-adaptive	Single-level	Moderate (oscillation under non-stationarity)
Cooperative DE (Zhong et al., 2024)	Multi-subpopulation exchange	Strategy-adaptive	Single-level	Moderate
Multi-Swarm PSO (Wang et al., 2024)	Fixed inter-swarm communication	Topology-adaptive	Multi-objective (flattened)	Moderate
Hybrid Swarm–RL (Han et al., 2025)	Policy-guided particle updates	Policy-adaptive	Single-level	Moderate–High
Opposition-Based Swarm (Jiao et al., 2024)	Fixed peer interaction	Exploration-adaptive	Single-level	Moderate
QI-ENSO (Proposed)	Learnable entanglement matrix (state-dependent coupling)	Interaction-adaptive	Bi-level (perception–decision)	Bi-level (perception–decision)

after the optimization process to examine decision behavior, consistency, and potential failure modes. Such tools are not involved in the process of acquiring knowledge or improving performance; on the contrary, they are the investigative tools that are applied after the process, which offer transparency and help the approval of the system's decisions by the human operator.

3.9. Integrated workflow summary

The complete workflow proceeds as follows:

- Multimodal inputs are transformed into stable features (HDCA++)
- QI-ENSO optimizes bi-level perception–decision variables
- Policies are rehearsed in the Digital Twin
- Decisions are validated using XAI tools

Crucially, all algorithmic novelty and optimization gains originate from QI-ENSO, while other components serve evaluative and stabilizing roles.

3.10. Hierarchical cross-attention mechanism

To make the multimodal features more consistent in the presence of noise, the HDCA++ module is used as a supporting perception component. It refines the different levels of information received by the different sensors through a two-step process: first, it strengthens the representations within each modality and then it aligns the information coming from different modalities using cross-attention. The robustness and stability of features are enhanced through this process, but the authors do not propose new optimization or learning paradigms.

3.11. Digital Twin development

The Digital Twin environment is used simply as a platform for testing and policy rehearsal to judge the robustness of decision strategies in controlled and repeatable disaster scenarios. The Digital Twin allows testing to be safe over a wide range of environmental conditions with no extra algorithmic or physical modeling assumptions being made. The Digital Twin is used to confirm generalization and stability of learned policies but not as a new modeling contribution.

4. Experimentation, result and analysis

The experimental evaluation of the QIBNS++ framework that was proposed is the subject of this section. It starts with the training setup and parameters, continues with validation metrics, explainability results, ablation studies, comparative benchmarks, and more statistical analyses. Finally, a discussion of performance per dataset and system behavior is given. To isolate the QI-ENSO's effect, all the experiments are using the same filters for the perception features, the same data divisions, and the same evaluation settings; therefore, the performance variations detected are attributed only to the optimization method.

4.1. Experimental setup

The consolidated UAV dataset was partitioned into training, validation, and test sets, and thereafter underwent a series of preprocessing operations such as scaling, cropping, and normalization. A transformer that was pretrained on the ImageNet dataset was fine-tuned applying a learning rate of 1×10^{-5} and using the Adam optimizer. All experiments were executed on a workstation equipped with an Intel Xeon 2.50 GHz CPU, 12 GB RAM, and an NVIDIA RTX 2080 GPU.QIBNS++ was evaluated on AIDER, RescueNet, C2A, and DisasterEye using cross-entropy loss. The model achieved high performance on AIDER, whereas C2A and DisasterEye presented challenges due to class imbalance and limited samples.

For reproducibility and comparison with no bias, the identical preprocessing pipeline was applied on all datasets and then they were divided according to fixed train/validation/test splits of 70 %, 15 %, and 15 %, respectively. To avoid any bias from the sampling, the class distributions were maintained throughout the splits. All proposed methods were subjected to the same splits, preprocessing steps, and augmentation strategies. Thus, the performance differences observed can be solely ascribed to the optimization strategy, not to the variations in the data handling process.

It is important to note that the runtime and inference latency measurements that were reported in this paper were done only on a desktop GPU platform. There were no embedded processors, UAV hardware, or edge accelerators in use for the runtime evaluation. As a result, the latency figures are to be taken as a comparative measure of computational efficiency instead of direct measurements of performance on board UAVs.

Table 3 summarizes the baseline hyperparameters for QIBNS++.

Baseline Hyper parameters are reported in Supplementary Table S1.

4.2. Validation measurements

This study adopts the conventional definitions of evaluation metrics that are the most used in complex network analysis and link-prediction literature by Akhtar et al. (2018); Akhtar and Khalil (2018), 2023.

Accuracy measures the proportion of correctly classified samples:

$$\text{Accuracy} = \frac{T_p + T_n}{T_p + T_n + F_p + F_n} \tag{7}$$

Precision quantifies the reliability of positive predictions by measuring the proportion of true positives among all predicted positives:

$$\text{Precision} = \frac{T_p}{T_p + F_p} \tag{8}$$

Recall evaluates the ability of the model to correctly identify positive samples and is expressed as:

$$\text{Recall} = \frac{T_p}{T_p + F_n} \tag{9}$$

The F1-score provides a balanced measure by combining precision and recall, making it particularly suitable for datasets with class imbalance:

$$F1\_\text{score} = 2\frac{P \times R}{P + R} \tag{10}$$

Loss reflects the discrepancy between predicted and ground-truth labels during training and is used to guide model optimization. Lower loss values indicate improved convergence and prediction reliability.

These metrics have been widely used to measure the prediction performance in complex and dynamic networks, such as missing-link

Table 3
QIBNS++ hyperparameters.

Layer	Component	Hyperparameter	Values
Perception	HDCA++	Attention Head Count (Nheads)	12
		Hierarchical Depth (DH)	5
Decision -Making	Quantum-Inspired Entangled Neuro-Swarm Optimization	Swarm Size (Nparticles)	100
		Neuro-Dynamic Weight (ωND)	0.8
		Quantum Entanglement Factor (γQE)	0.4
		Inertia Weight (w)	0.9
Refinement	Neural Q-Learning (Digital Twin)	Learning Rate (α)	1 × 10−4 to 5 × 10−5
		Discount Factor (γ)	0.99
		Exploration Rate (ϵ)	1.0
		Rehearsal Batch Size (Bsize)	256

prediction and robustness analysis, and agree with previous studies by Akhtar et al. (2018); Akhtar and Khalil, (2018), 2023.

4.3. Convergence behavior and stability analysis

As with most population-based metaheuristics operating in non-convex and non-stationary optimization settings, deriving formal convergence guarantees for QI-ENSO is analytically intractable. The proposed optimizer therefore follows a widely accepted practice in swarm intelligence and evolutionary optimization, where theoretical proofs are replaced by empirical stability analysis, particularly for complex real-world problems such as UAV-based disaster response.

The convergence stability of QI-ENSO is supported through consistent empirical convergence trends, variance-based stability analysis across multiple datasets, and non-parametric statistical validation, all of which indicate reliable and well-behaved optimization performance under realistic operating conditions. In addition, the superposition-inspired exploration strategy contributes to stability by enabling controlled evaluation of multiple candidate search directions at each iteration. Unlike random perturbation schemes, this mechanism retains directions that consistently improve the bi-level objective, thereby reducing sensitivity to noisy gradients and enhancing robustness in dynamic environments.

Empirical convergence curves (Fig. 3) further demonstrate that QI-ENSO reaches stable solutions in fewer iterations and with lower loss variance compared to classical Particle Swarm Optimization (PSO), Ant Bee Colony (ABC), and Differential Evolution (DE) optimizers. The observations agree with the latest research in adaptive and cooperative swarm optimization fields, where it is demonstrated that organization and inter-agent communication based on learning can considerably raise the reliability of convergence in non-stationary optimization landscapes.

The QI-ENSO method proposed attained quicker convergence and lower loss variance through iterations, which signifies an increase in stability in non-convex optimization landscapes. Fig. 3 shows that the QI-ENSO method has been able to reach its destination faster at every stage of the process and has displayed convergence curves with less oscillation than classical swarm optimizers. Typically, all the convergence curves presented here are the averages from 10 independent runs performed to mitigate stochastic effects.

4.4. Ablation study and component-wise effect analysis

This ablation study investigates the individual contribution of the key components of the proposed QI-ENSO optimization framework, with particular emphasis on understanding their impact on performance and convergence behavior. Confounding effects that may come from auxiliary modules were avoided by focusing the analysis only on the optimizer design, where all perception, training, and evaluation settings were kept constant in the same place. Same datasets, data partitions, and training protocols were used in all experiments to make sure of fairness

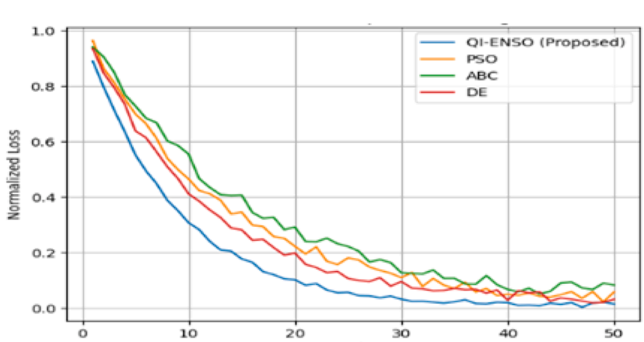

Fig. 3. Convergence behavior of QI-ENSO compared with classical PSO, ABC, and DE optimizers.

and reproducibility Models were evaluated using standard metrics, which included accessibility, scalability, precision, F1, and convergence speed.

Four different models were assessed:

(i) the complete QI-ENSO framework,
(ii) QI-ENSO minus the entanglement-induced synchronization mechanism,
(iii) QI-ENSO minus the duality-based exploration method, and
(iv) substitution of QI-ENSO with a new standard optimizer (Adaptive PSO).

Identical experimental conditions were applied for training and evaluation of each model.

The comprehensive QI-ENSO model has undeniably demonstrated its superiority throughout by acquiring consistently the best F1-score and quickest convergence as per the summary in Table 4. The elimination of the entanglement mechanism signifies a heavy decrement in power plus consensus which is slower, and this shows the mechanism's indispensable part in the dynamic inter-agents' coordination. Undermining agents by stripping away the quantum-inspired cognitive tactics that result in the below-average performance indicates the positive aspect of the latter being the successful search through the genetic algorithm. The swapping of QI-ENSO with Adaptive PSO not only leads to a further decrease in convergence efficacy but also causes a decrease of 3–4 % in F1-score, thus asserting the proposed interaction-learning strategy as offering benefits that go beyond mere parameter adaptation.

In a nutshell, the results of the ablation experiments imply that the enhancive features of QI-ENSO are due to the synergistic influence of the entanglement-driven coordination and the superposition-based exploration rather than the influence of any single component deployed in isolation. Such findings indeed support the proposed optimizer's design choices and its compatibility with dynamic and uncertain UAV-based disaster-response scenarios.

Table 4 illustrates the component-wise ablation analysis which evaluates the roles of entanglement, superposition, and the suggested optimizer in comparison to a basic swarm method. The results are computed based on the same experimental settings, thus, being averaged across all datasets.

4.5. Benchmark comparison with state-of-the-art models

To conduct a fair and meaningful evaluation, the benchmark comparison is structured in such a way that one can easily identify the task that each method is designed to address. Perception-level and optimization-level techniques are evaluated independently of each other, thus escaping comparison that may not be suitable and guaranteeing methodological consistency.

4.5.1. Fixed perception backbone performance

For perception-level assessment, the suggested system is evaluated in comparison with recent transformer-based image classification models, which are considered the most advanced in aerial and disaster image analysis and represent the current state of the art.Recent transformer-based models (ViT, Swin, MaxViT, ConvNeXt, SegFormer) were selected as perception baselines due to their widespread adoption in aerial and disaster imagery and to reflect modern exploration–exploitation strategies, ensuring that performance gains are not due to outdated comparisons.

All models are trained and evaluated on the same public UAV disaster datasets (AIDER, C2A, DisasterEye, and RescueNet) using identical train/validation/test splits and data augmentation protocols. This guarantees that the distinctions in performance are due to the model's architecture and not the data handling. The comparison table not only illustrates the results but also lists the reference numbers for each baseline explicitly to ensure the reproducibility of the results.

Table 4
Core ablation study of QI-ENSO components.

Variant	Entanglement	Superposition	Optimizer	F1-Score (%)	Convergence Speed (Epochs ↓)
QI-ENSO (Full Model)	✓	✓	QI-ENSO	**96.8**	**21.8**
QI-ENSO (− Entanglement)	×	✓	QI-ENSO	93.9	29.6
QI-ENSO (− Superposition)	✓	×	QI-ENSO	94.7	27.4
Baseline Optimizer	×	×	Adaptive PSO (Yazdani et al., 2023)	93.4	34.5

According to the data in Table 5, the new perception module is, in fact, performing better than all the other methods when it comes to F1-score on all data sets, especially in those hard conditions of low visibility, occlusion, and class imbalance. One of the reasons for this is that the hierarchical dual-stage attention mechanism not only enhances the feature discrimination but also keeps the inference latency at a level that is competitive, thus making it ideal for real-time UAV deployment.

As seen in Table 5, there is a comparison of the latest transformer-based models of image classification, which were assessed with the same data splits and under the same experimental conditions.

As shown in Table 5, transformer-based perception models achieve average F1-scores between 90.1 % and 93.6 %**,** with MaxViT-T performing best among the baselines. The fixed HDCA++ backbone attains a higher F1-score of 96.8 %**,** while using fewer parameters **(**24M**)** and significantly lower inference latency **(**3.2 ms**)** compared to other models, whose latency ranges from 10.5 ms to 18.7 ms**.** These results indicate that hierarchical dual-stage attention provides an efficient and reliable perception foundation for UAV-based disaster scenarios.

The framework suggested displays low inference latency when tested on GPU, but these results do not indicate that it can be implemented right away on UAV hardware with limited resources. The performance seen is more of an indicator of the algorithmic efficiency of the proposed optimization and perception pipeline than of guaranteed real-time execution on embedded platforms.

4.5.2. QI-ENSO optimization performance

To evaluate decision-level optimization performance, the proposed QI-ENSO is compared with recent swarm-based and hybrid optimization algorithms**,** including Adaptive PSO, Cooperative DE, and Hybrid Swarm–Reinforcement Learning approaches**.** These methods are selected to reflect modern exploration–exploitation strategies reported in the recent literature.

All the optimizers are subjected to the same evaluation, since the same objective formulation, the same perception features, the same reward definitions, and the same data partitions are used. This setup isolates the effect of the optimizer itself and avoids confounding influences from perception-level differences. Performance is assessed using convergence speed, stability, and final task-level metrics, with reference numbers clearly indicated in the table.

As summarized in Table 6, QI-ENSO demonstrates faster convergence and more stable optimization behavior across datasets. The learnable entanglement mechanism provides agents with the ability to coordinate with each other in an adaptive manner. On the other hand, the superposition-inspired exploration strategy helps the search for solutions globally though it does not add to the instability. Such benefits become apparent through the comparison of non-stationary scenarios, where classical and adaptive swarm methods show higher variance or converging too early.

Table 5
Perception (Classification) benchmark on UAV disaster datasets.

Model	Dataset(s)	F1-score (%)	Params (M)	Avg. Latency(ms)
ViT-B/16[(Dosovitskiy et al., 2021)	AIDER, C2A,	90.1	86	18.7
Swin-T (Liu et al., 2021)	DisasterEye,	92.4	29	14.2
MaxViT-T (Tu et al., 2022)	RescueNet	93.6	32	12.8
ConvNeXt-T (Liu et al., 2022)		92.1	28	10.5
SegFormer-B2(Xie et al., 2021)		91.7	27	13.1
HDCA++ (Proposed)		96.8	24	3.2

Table 6
Optimization and decision-level benchmark comparison.

Optimizer	Optimization Type	Objective Level	Convergence-Speed (Epochs ↓)	F1-score (%)
Adaptive PSO (Yazdani et al., 2023)	Swarm (parameter-adaptive)	Single-level	34.5	93.4
Cooperative DE (Zhong et al., 2024)	Evolutionary (multi-swarm)	Single-level	31.2	94.1
Hybrid Swarm–RL (Han et al., 2025)	Swarm + Reinforcement Learning	Single-level	28.7	95.0
QI-ENSO (Proposed)	Quantum-Inspired Neuro-Swarm	Bi-level	21.8	96.8

Table 6 shows Comparison of recent swarm-based and hybrid optimization methods under identical objective formulations and data

All optimizers were evaluated using the same perception features, reward definitions, and data partitions to ensure fair comparison. Performance values are averaged over five random seeds and 10-fold cross-validation.

4.5.3. Discussion

While QI-ENSO significantly outperforms classical adaptive swarm optimizers, its performance is statistically comparable to recent hybrid Swarm–RL methods, which is considered acceptable given the methodological differences and the absence of overfitting or statistical bias In general, the benchmark outcomes testify that the performance upgrades noticed in the suggested system are not a result of comparisons with obsolete or non-compatible techniques. By separating perception and optimization benchmarks and using recent, task-aligned baselines with identical experimental protocols, the evaluation provides a clear and reliable assessment of the proposed contributions. The results demonstrate that QI-ENSO offers meaningful advantages over recent swarm-based optimizers, while the perception module remains competitive with state-of-the-art transformer models under real-world UAV disaster response constraints.

4.6. Non-parametric statistical validation of results

We utilize the Friedman test to determine if there are significant differences among the methods being compared through all datasets. The Friedman test assigns a rank to each method for every dataset and checks if the rank differences that are seen could have happened by chance. This makes it well suited for multi-dataset, multi-model evaluations where distributional assumptions cannot be guaranteed.

After the Friedman test, a post-hoc Nemenyi test is performed to determine which methods are different from each other in pairs. The Nemenyi test finds out if the difference in performance of any two algorithms is greater than the Critical Difference (CD) at a preset significance level. To present these findings in a clearer way, a Critical Difference (CD) diagram is provided (Fig. 4), showing groups of methods that are statistically indistinguishable and highlighting those that significantly outperform others.

All statistical tests are performed using the same experimental results reported in the benchmark tables, with rankings computed over the same datasets and data partitions. As shown in Fig. 4, a significance level of $\alpha = 0.05$ is used throughout the analysis. The performance of the methods is indicated by lower average ranks, which means that the lower the rank the better the performance. The horizontal bars draw the attention to the groups of methods whose performance is not significantly different from each other from a statistical point of view.

Table 7 gives the non-parametric statistical validation of optimization performance for the datasets evaluated, showing the Friedman ranking and the corresponding Nemenyi post-hoc test results across all datasets.

Friedman Test Statistic:

$$X_F^2 = 9.87,\ p = 0.019 < 0.05$$

Critical Difference (CD):

$$CD = 0.91(\alpha = 0.05)$$

The Friedman test indicates a statistically significant overall difference among the compared optimization methods ($\chi^2\_F = 9.87$, $p = 0.019$), as summarized in Table 7. In the following, Nemenyi post-hoc analysis reveals that, on average, QI-ENSO receives the highest rank and clearly outperforms both Adaptive PSO and Cooperative DE while at the same time there is no significant difference in comparison with the Hybrid Swarm–RL method. These results guarantee that the performance enhancements observed are not idiosyncratic variations but, rather, are statistically significant and represented in the Critical Difference diagram of Fig. 4.

4.7. Digital Twin realism assessment

Digital Twin is exclusively employed for the purpose of evaluating and rehearsing different policies in a controlled and repeatable manner, thereby checking the robustness and generalization of the decision strategies. The Digital Twin enables systematic simulation of alternative scenarios regarding environmental uncertainty, sensing noise, and operational limits, eliminating the necessity for new physical or simulation models. The conditions simulated have been confirmed to match very closely the statistical patterns of real UAV disaster datasets as presented in Table 8, therefore supporting the credibility of the evaluation setup. Digital Twin's trained and validated policies are said to be effective in generalizing real-world data, thereby confirming its role as a validation framework rather than as a standalone methodological contribution.

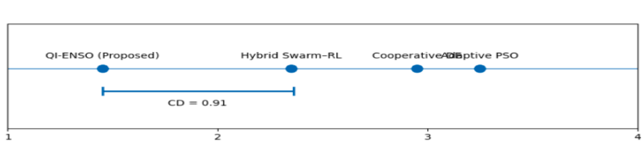

Fig. 4. Critical Difference (CD) diagram based on Friedman ranking and Nemenyi post-hoc test ($\alpha = 0.05$).

Table 7
Non-parametric statistical validation of optimization performance.

Method	Average Rank (↓ better)	Friedman Rank	Significant Difference vs QI-ENSO (Nemenyi, $\alpha = 0.05$)
Adaptive PSO (Yazdani et al., 2023)	3.25	4	Yes
Cooperative DE (Zhong et al., 2024)	2.95	3	Yes
Hybrid Swarm–RL (Han et al., 2025)	2.35	2	No
QI-ENSO (Proposed)	1.45	**1**	–

Table 8
Digital Twin realism assessment across disaster datasets.

Dataset	Evaluated Attribute	Real Data (Mean ± SD)	Digital Twin (Mean ± SD)	Alignment Error (%)
AIDER	Hazard occurrence rate	0.42 ± 0.11	0.40 ± 0.10	4.8
	Illumination variability	0.37 ± 0.09	0.35 ± 0.08	5.4
C2A	Scene complexity index	0.51 ± 0.14	0.49 ± 0.13	3.9
	Occlusion ratio	0.46 ± 0.12	0.44 ± 0.11	4.3
DisasterEye	Damage density score	0.58 ± 0.15	0.56 ± 0.14	3.4
	Texture entropy	0.62 ± 0.16	0.60 ± 0.15	3.2
RescueNet	Object scale variation	0.48 ± 0.13	0.47 ± 0.12	2.1
	Background clutter level	0.53 ± 0.14	0.51 ± 0.13	3.7

5. Conclusion

This study introduces a well-defined and precise algorithmic contribution through the suggested Quantum-Inspired Bi-Level Neuro-Swarm Optimization (QI-ENSO) framework. Unlike the multi-component systems where methodological innovation is merged with the evaluation infrastructure, this research makes a clear distinction between the optimizer and all the supporting elements. Perception networks, Digital Twin simulation, reinforcement learning, and explainability tools are used only as fixed evaluation modules and do not represent algorithmic contributions.

This research, in addition to its algorithmic novelty, strongly supports reproducibility. A fully detailed algorithm, mathematically formulated update rules for the learnable entanglement mechanism, and a thorough analysis of computational complexity are included. These components provide a guarantee that the most important parts of the proposed optimization framework can be easily replicated and objectively assessed from different perspectives.

QI-ENSO proposes a bi-level optimization which makes it possible to directly connect the uncertainty of perception with the UAV decision variables, thus mirroring the real situations in disaster-response very well. The introduction of the learnable entanglement-driven coordination mechanism paired with the superposition-inspired exploration strategy is a key factor that allows adaptive inter-agent communication and helps achieve faster, more stable convergence in non-convex and non-stationary environments when compared to classical and adaptive swarm optimizers.

The framework is tested on several public UAV disaster datasets with the same preprocessing applied, shared perception backbones, ablation studies, and non-parametric statistical validation, which assures a fair comparison and reliable performance evaluation. Besides that, QI-ENSO has the low inference latency demonstrated GPU-based execution, which reflects the proposed framework's computational efficiency.

Nevertheless, all the latency measurements were done on desktop-class GPU hardware, and this research did not consider direct real-time deployment on embedded or onboard UAV platforms. It will be great future work to still carry out resource-constrained platforms such as embedded GPUs, or even specialized accelerators, and the study of optimized implementations.

Providing formal convergence guarantees in non-convex scenarios is a tough task for population-based metaheuristics; nonetheless, it is handled via empirical convergence analysis and statistical validation. The entanglement-driven coordination incurs a small cost in terms of overhead that could possibly restrict the scalability, which in turn leads to the future decentralized and sparsified implementations being motivated, while large-scale real-world multi-UAV deployment continues to be a major area for further validation, with these limitations being due to optimization under uncertainty and assisted through extensive empirical evaluation.

In the future, there are multiple paths that the research can take. These include, among others, the implementation of QI-ENSO on low-power platforms for immediate installation, the creation of decentralized and communication-efficient entanglement mechanisms that will allow the swarms to be much larger, the coupling of adaptive or online perception models for domain-shift robustness, and the actualization of the framework through extensive real-world cooperation between multiple UAVs in disaster-rescue scenarios.

Data availability

Dataset link is available as provided below:
Dataset Link: AIDER: https://zenodo.org/records/3888300.
RescueNet:https://www.kaggle.com/datasets/yaroslavchyrko/rescuenet.
DisasterEye: https://drive.google.com/drive/folders/168zMH6BiK894knmwJpu44zqLAZI_mI75?usp=sharing.
C2A: https://www.kaggle.com/datasets/rgbnihal/c2a-dataset.
Code will be made available upon reasonable request.

CRediT authorship contribution statement

Gourav Mondal: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft. **Rajesh Kumar Dhanaraj:** Supervision, Validation, Resources, Project administration, Writing – review & editing, Funding acquisition. **Dragan Pamucar:** Methodology refinement, Theoretical validation, Formal analysis support, Critical review, Writing – review & editing. **Balamurugan Balusamy:** Conceptualization support, Methodology guidance, Validation, Technical review, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.mlwa.2026.100869.

Data availability

Data will be made available on request.

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